



Summary: AI research scientist, interested in building autonomous agents and cool tools for scientific discovery, with deep expertise in LLM-based reasoning, Bayesian optimization, and surrogate modeling applied to materials inverse design and drug discovery.



Education



Doctor of Philosophy in Chemical Engineering

(May 2023)

Master of Science in Chemical Engineering

(Oct 2021)

University of Rochester, Rochester, NY

Thesis: "Applications of Physics-informed Machine Learning in Chemical Engineering". Advisor: [Andrew D White](#)

Master of Science in Environmental Engineering

(May 2018)

Missouri University of Science and Technology (UMR), Rolla, MO

Thesis: "Numerical Modeling of Capillary-driven Flow in Open Microchannels: An Implication of Optimized Wicking Fabric Design"

Bachelor of Science in Chemical Engineering

(July 2015)

University of Tehran, Iran

Thesis: "Experimental Setup and Optimization for Electro-catalytical Generation of Hydroxyl Radicals in Wastewater Treatment"



Work Experience



Senior Scientist

Lila Sciences, Boston, MA

(June 2025 – present)

- Designed scientific AI agents with memory management for long-horizon inverse molecular design using LLM reasoning, and surrogate modeling
- Built stateful code execution environments for autonomous experimentation, data processing, and surrogate model training
- Developed a scalable multi-modal RAG system over scientific literature, benchmarked against PaperQA2/LitQA2
- Built a research agent that autonomously reproduces and improves PhD-level manuscripts without human intervention
- Led and mentored cross-functional teams of PhD-level researchers and engineers on scientific AI projects in materials discovery

Staff Scientist

(Jan 2024 – June 2025)

Acceleration Consortium Research Fellow

Acceleration Consortium, Toronto, ON

(June 2023 – Dec 2023)

- Applied representation learning, Bayesian optimization, and AI agents to accelerate self-driving chemistry lab campaigns
- Incorporating built-in domain knowledge in self-driving chemistry lab workflows, materials inverse design problem and hit-to-lead optimization
- Developed [chemistry-informed AI agents](#) within Slack API
- Developed [agent-based chemistry plugins](#) via large language models to accelerate design of materials for CO₂ reduction
- Mentored undergraduate and graduate-level students on applied AI in computer-science and materials discovery projects
- Co-organized global hackathons for application of AI in chemistry and materials

Research Assistant

University of Rochester, Rochester, NY

(2019 – 2023)

- Developed a [HuggingFace app](#) that answers questions from scientific papers using OpenAI's large language models
- Developed a [edge-computing cheminformatics tool](#) for semi-supervised classification of the activity of antimicrobial peptides via positive-unlabeled learning using recurrent neural networks
- Developed a [disease modeling tool](#) to predict future disease spreads and infer location of patient-zero
- Developed an [automated tool](#) in CFD modeling that reduces the number of simulations using active learning and generates a symbolic equation for the system of interest via symbolic regression
- Contributed to development of a [simulation-based inference tool](#) via maximum entropy reweighting
- Contributed to development of a plugin with TensorFlow GPU-accelerated operations combined with HOOMD-Blue molecular dynamics simulation engine ([HOOMD-TF](#))
- Developed a [web-app](#) for peptide-based gelator transparency classification using Kernel ridge regression
- Developed an [automated tool](#) on a Raspberry-Pi for real-time monitoring of HPC using Python, JS and HTML
- Implemented finite difference analysis in Python to study [2D shallow water dynamics](#)
- Implemented Monte Carlo simulations in MATLAB to study evolution of spin configurations of a ferromagnet using the Ising model

Energy & Materials ML Research Engineer

(Aug 2022 – Mar 2023)

Energy & Materials Intern

Toyota Research Institute, Los Altos, CA

(May 2022 – Aug 2022)

- Developed a deep learning architectures to predict degradation of used Li-ion batteries with unknown cycling histories ([US Patent pending](#)).

Manufacturing Process Modeling Intern

The Goodyear Tire & Rubber Company, Akron, OH

(May-Dec 2017)

- Phase-change heat transfer modeling and optimization of tire vulcanization process in ANSYS
- Model verification based on plant data and analytical solution
- Utilized assets more efficiently through MATLAB post processing and automating the simulation process using OPTIMUS
- Provided faster simulation results using Adaptive Mesh Refinement and High-Performance Computing
- GUI development and coupling ANSYS with MATLAB for time-effective post processing

Research Assistant

Missouri University of Science and Technology, Rolla, MO

(2016 – 2018)

- Numerical modeling of multiphase flow in open microfluidics using ANSYS and STAR-CCM + with by developing adaptive mesh refinement algorithms

Engineering Intern

Emden-Leer University of Applied Sciences, Emden, Germany

(July-Sept 2014)

- Design of experiments in advanced oxidation process (AOP) for wastewater treatment



Computer Skills



Scientific Softwares Developed:

- [MaxEnt](#)
- [Py0](#)
- [HOOMD-TF](#)
- [Decode-ELM](#)
- [Eunomia](#)
- [AL-CFD](#)
- [Peptide.bio](#)
- [dZiner](#)
- [Bye-Cycle](#)
- [Heisenberg](#)

Other tools: TensorFlow, PyTorch, Scikit-learn, JAX, LiteLLM

Infrastructure: HPC Slurm, Ray, Kubectl, Lambda labs, AWS

Languages: Python, JavaScript, HTML, CSS



Honors and Awards



3 rd Place Radical AI Winner at the 2024 LLM Hackathon - Materials and Chemistry Toronto, ON	(May 2024)
2 nd Place Winner Team at the Bayesian Optimization Hackathon for Chemistry and Materials Toronto, ON	(Apr 2024)
Acceleration Consortium Research Fellowship University of Toronto, Toronto	(May 2023)
1 st place winner at the Battery Informatics & ML Kaggle Competition Materials Research Society, Boston, MA	(Dec 2022)
Kwang-Yu and Lee-Chien Wang Fellowship Department of Chemical Engineering, University of Rochester	(Nov 2021)
Earl W. Costich Graduate Fellowship Department of Chemical Engineering, University of Rochester	(May 2020)
1 st place winner: 2017 Mike Alizadeh Scholarship American Society of Civil Engineers (ASCE)	(Aug 2017)
Recognized reviewer: <i>Journal of Environmental Chemical Engineering</i>	(May 2016)



Selected Publications and Patents ([Google Scholar](#))



1. 32 examples of LLM applications in materials science and chemistry <i>Machine Learning in Science & Technology (Project Co-organizer and team lead)</i> Y Zimmermann, A Bazgir, A Al-Feghali, M Ansari , and others	(Sept 2025)
2. Kernel learning assisted synthesis condition exploration for ternary spinel <i>Nature Communication Materials (Project Co-Lead)</i> Y Liu, M Ansari , R Black, J Hattrick-Simpers	(July 2025)
3. Bayesian optimization hackathon for chemistry and materials <i>chemRxiv (Project Co-Lead)</i> S Baird, M Ansari , Z Afzal, Q Ai and others	(June 2025)
4. dZiner: Rational Inverse Design of Materials with AI Agents <i>Spotlight AI4Mat-NeurIPS 2024 (Project Lead)</i> M Ansari , J Watchorn, CE Brown, JS Brown	(Oct 2024)
5. Agent-based Learning of Materials Datasets from Scientific Literature <i>Digital Discovery</i> M Ansari , SM Moosavi	(Oct 2024)
6. Learning Peptide Properties with Positive Examples Only <i>Digital Discovery</i> M Ansari , AD White	(Mar 2024)
7. History-agnostic Battery Degradation Inference and US Patent Pending <i>Journal of Energy Storage</i> M Ansari , S Torrisi, A Trewartha, S Sun	(Mar 2024)
8. Serverless Prediction of Peptide Properties with Recurrent Neural Networks <i>Journal of Chemical Information and Modeling</i> M Ansari , AD White	(Apr 2023)
9. Assessment of Chemistry Knowledge in Large Language Models that Generate Code <i>Digital Discovery</i> AD White, GM Hocky, HA Gandhi, M Ansari , S Cox, GP Wellawatte, S Sasmal, Z Yang, K Liu, Y Singh, WJ Peña Ccoa	(Jan 2023)
10. Book chapter: Hyper-parameter Optimization in Deep Learning <i>Deep Learning for Molecules and Materials, Living Journal of Computational Molecular Science</i> M Ansari , AD White	(July 2022)
11. Inferring Spatial Source of Disease Outbreaks using Maximum Entropy <i>American Physical Society, Physical Review E</i> M Ansari , D Soriano-Paños, G Ghoshal, AD White	(July 2022)
12. Iterative Symbolic Regression for Learning Transport Equations <i>AIChE Journal, Special Edition for AI</i> M Ansari , HA Gandhi, DG Foster, AD White	(Mar 2022)
13. Simulation-based Inference with Approximately Correct Parameters via Maximum Entropy <i>NeurIPS 2020, Machine Learning in Science and Technology</i> R Barrett, M Ansari , G Ghoshal, AD White	(Apr 2022)



Presentations and Talks



1. Teaching computers how to do science (Invited talk) IBM Research, Tokyo, Japan	(Feb 2025)
2. Rational Inverse Design of Materials with Chemist AI Agents <i>Spotlight AI4Mat-NeurIPS 2024, Vancouver, BC</i>	(Dec 2024)
3. Multi-modal AI Agents in Materials Discovery <i>Accelerate 24, Vancouver, BC</i>	(Aug 2024)
4. Materials Inverse Design with AI Agents (Invited talk) Trillion Parameter Consortium, Barcelona, Spain	(June 2024)
5. AI That Probably Wouldn't Kill You (Invited talk) 2024 LLM Hackathon - Materials and Chemistry, Toronto, ON	(May 2024)
6. Flexible Automation of Self-driving Labs with Built-in Domain Knowledge (Invited talk) Toyota Research Institute, Los Altos, CA	(Mar 2024)
7. Positive Unlabeled Learning of Peptide Properties <i>Accelerate23 Conference, Toronto, ON</i>	(Aug 2023)
8. Maximum Entropy Inference in Chemical Reaction Networks with Unknown Kinetic Parameters <i>AIChE, Phoenix, AZ</i>	(Nov 2022)
9. Rescuing Physics-based Models with Maximum Entropy Reweighting <i>Wang Lecture, University of Rochester, NY</i>	(Sept 2022)
10. Serverless Prediction of Peptide Properties with Recurrent Neural Networks <i>Middle Atlantic Regional Meeting of the American Chemical Society, Ewing Township, NJ</i>	(June 2022)
11. Simulation-based Inference with Approximately Correct Parameters via Maximum Entropy <i>NeurIPS 2020: Workshop on Machine Learning for Structural Biology</i>	(Dec 2020)